EXPERIMENT NUMBER: 1

Simulation and Comparative Analysis of FDMA, TDMA and CDMA: Resource Allocation, User Capacity and Mobile-Generation Context

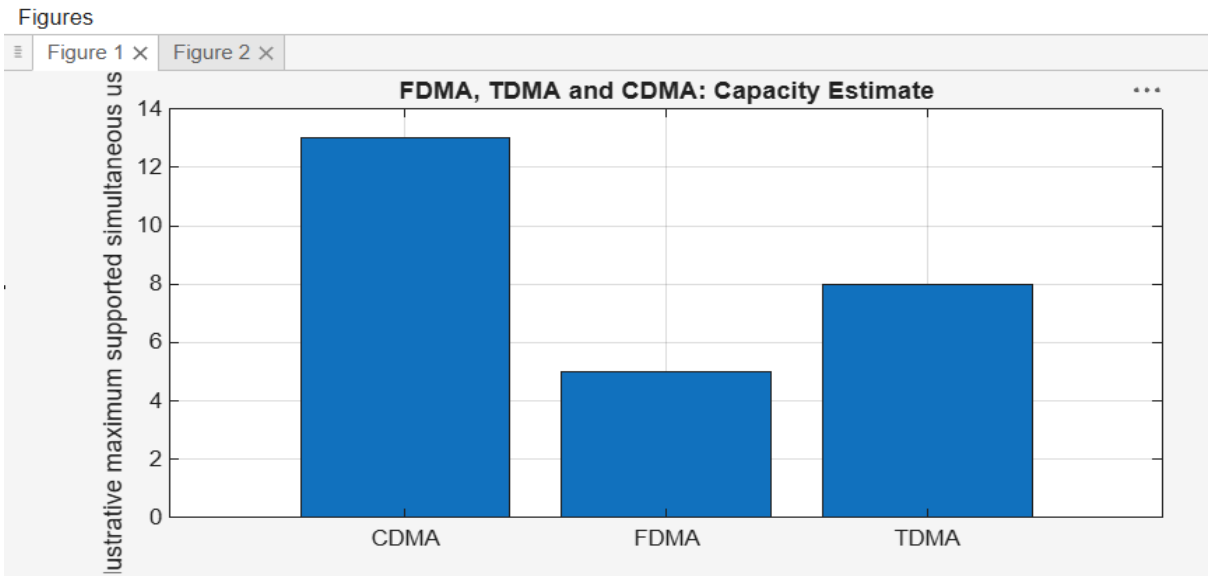
OBJECTIVE 1:

Matlab Program

```
% Simplified illustrative model; not a full cellular-network simulation.
clear; clc; close all;
% Illustrative system parameters
totalBandwidthHz = 1e6;
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7;
% Capacity estimates under the stated assumptions
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
cdmaUsers, targetSIRdB);
% CDMA SIR as the number of simultaneous users increases
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
% Display a comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
```

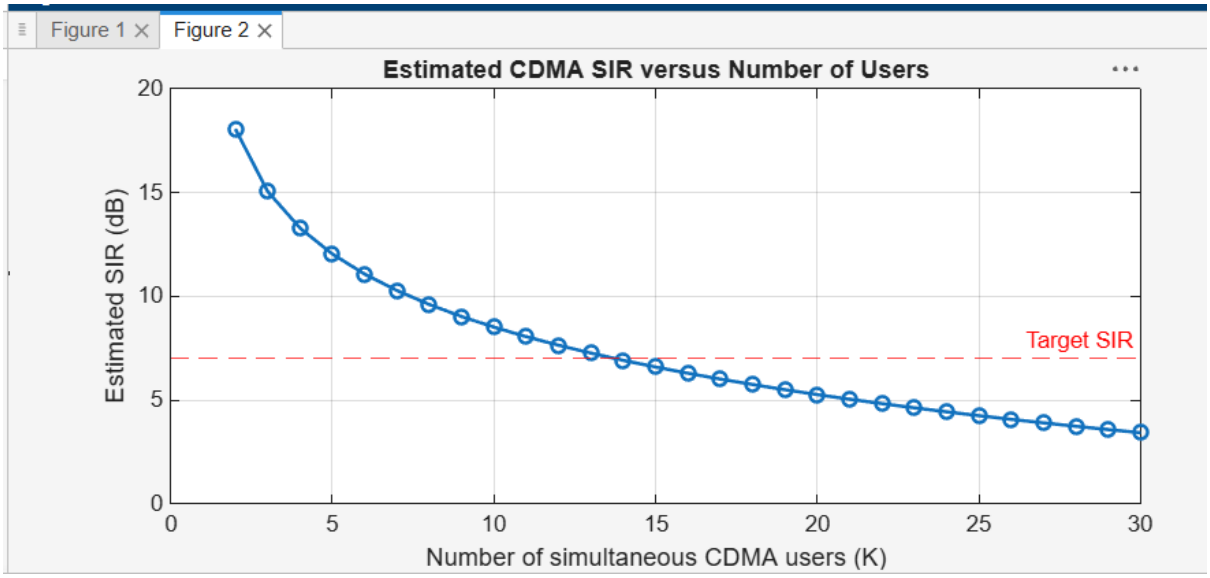
```
results = table(method, supportedUsers);
disp(results);
% Plot estimated user capacity
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
% Plot estimated CDMA SIR as users are added
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

Result



OBJECTIVE 2:

Result



OBJECTIVE 3:

Matlab Program

```
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];

system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];

accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];

keyIdea = ["Separate frequency channels"; ...
    "Separate time slots"; ...
    "Code-based user separation"; ...
    "Wideband code-based access"; ...
    "Orthogonal frequency resources"; ...
    "Flexible OFDM-based radio access"];

evolutionTable = table (generation, system, accessMethod, keyIdea);

disp(evolutionTable);
```

Result

```
Command Window

>> Exp1slotA
    generation    system    accessMethod              keyIdea
    __________    ______    ____________    ____________________________________

       "1G"       "AMPS"    "FDMA"          "Separate frequency channels"
       "2G"       "GSM"     "TDMA"          "Separate time slots"
       "2G"       "IS-95"   "CDMA"          "Code-based user separation"
       "3G"       "UMTS"    "WCDMA"         "Wideband code-based access"
       "4G"       "LTE"     "OFDMA"         "Orthogonal frequency resources"
       "5G"       "5G NR"   "OFDMA-based"   "Flexible OFDM-based radio access"

>>
```

OBJECTIVE 4:

Matlab Program

```
figure;
tiledlayout(1,3);
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
    rectangle('Position',[0, u-1, 1, 0.8], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(0.5, u-0.6, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 1]);
ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
```

```
set(gca,'YTick',[]);
hold off;
% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[u-1, 0, 0.8, 1], ...
        'FaceColor',colors(u,:), 'EdgeColor','k');
    text(u-0.6, 0.5, sprintf('User %d',u), ...
        'HorizontalAlignment','center');
end
xlim([0 4]);
ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
% CDMA: users share the same time and frequency band using different codes
nexttile;
hold on;
for u = 1:4
    rectangle('Position',[0, 0, 1, 1], ...
        'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
        'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
```

```
xlim([0 1]);
ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
title('CDMA');
set(gca,'YTick',[]);
hold off;
sgtitle('Conceptual Time-Frequency Resource Allocation');
```

Result

